# Measures of Dispersion

Range, interquartile range, mean absolute deviation, variance, standard deviation and the coefficient of variation.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · #KAUMeLON · Official revision sheet

**KAU Me-LON** · www.agrinoteskau.com · Dept. of Agricultural Statistics, College of Agriculture, Vellayani

## 1 What is dispersion?

The **extent to which observations deviate from a central value**: the spread of the data.

| Absolute measures | Relative measures |
|---|---|
| In the data's units: range, IQR, MAD, SD | Unit-free: coefficient of variation (CV) |

A good measure is rigidly defined, easy to compute, based on all values, suitable for further algebra and not unduly affected by extremes.

## 2 Range & IQR

Range

$$R = L - S$$

Interquartile range

$$IQR = Q_3 - Q_1$$

- **Range**: simplest; uses only the two extremes.
- **IQR**: spread of the **middle 50%**; less affected by extremes.
- **Quartile deviation** = IQR ÷ 2.

## 3 Mean absolute deviation

MAD about the mean

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

Uses all observations but ignores signs. MAD is **least about the median**.

## 4 Variance & standard deviation

Sample variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Shortcut

$$s^2 = \frac{1}{n-1}\left\{\sum x_i^2 - \frac{(\sum x_i)^2}{n}\right\}$$

Discrete frequency

$$s^2 = \frac{1}{n-1}\left\{\sum f_i x_i^2 - \frac{(\sum f_i x_i)^2}{n}\right\}$$

Grouped, $d_i = (x_i - A)/c$

$$s = c\sqrt{\frac{1}{n-1}\left\{\sum f_i d_i^2 - \frac{(\sum f_i d_i)^2}{n}\right\}}$$

SD $s$ = non-negative square root of $s^2$. Grouped data: $x_i$ = class mark, $A$ = class mark with the highest frequency, $c$ = class interval.

## 5 Properties to remember

- ✓ Variance is in **squared** units; SD is in the **original** units.
- ✓ SD is always ≥ 0; SD = 0 means all observations are identical.
- ✓ Sample variance uses divisor $n - 1$ (unbiased).
- ✓ Adding a constant to every value does **not** change SD.
- ✓ Multiplying by $b$ multiplies SD by $|b|$.
- ✓ Squaring gives extreme deviations more weight.

## 6 Coefficient of variation

CV (%)

$$CV = \frac{s}{\bar{x}} \times 100$$

A **relative** measure, used to compare variability of datasets in **different units or with different means**.

| Higher CV | Lower CV |
|---|---|
| More relative variability: less consistent | Less relative variability: more consistent |

## 7 Comparing the measures

| Measure | Uses | Extreme values | Note |
|---|---|---|---|
| **Range** | 2 extremes only | Strongly affected | Quick, rough |
| **IQR** | Middle 50% | Resistant | Pairs with the median |
| **MAD** | All values | Moderate | Ignores signs |
| **SD** | All values | Sensitive (squared) | Best for algebra |
| **CV** | SD and mean | Sensitive | Unit-free; needs a positive mean |

## ! Common mistakes

- × Dividing by $n$ for a sample variance.
- × Reporting variance in the original units.
- × Comparing SDs of data in different units (use CV).
- × Forgetting to multiply by $c$ in the grouped shortcut.

## ? Fill in the blanks

1. Variety A has SD 4 q and mean 40 q; variety B has SD 3 q and mean 20 q. The more consistent variety has CV = _____ %. [ 10 (variety A) ]
2. If every observation is doubled, the SD is multiplied by _____ and the variance by _____. [ 2; 4 ]
3. If yield is recorded in kg, its variance is expressed in _____. [ kg² ]
4. The observations 5, 5, 5, 5 have a standard deviation of _____. [ 0 ]
5. With $Q_1$ = 18 and $Q_3$ = 30, the quartile deviation is _____. [ 6 ]
6. Adding 10 kg to every observation changes the SD by _____. [ nothing (0) ]

Source: *Textbook of Agricultural Statistics*, Ch. 6 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

© 2026 Department of Agricultural Statistics, College of Agriculture, Vellayani, KAU · www.agrinoteskau.com · Personal study only; no redistribution or AI use.